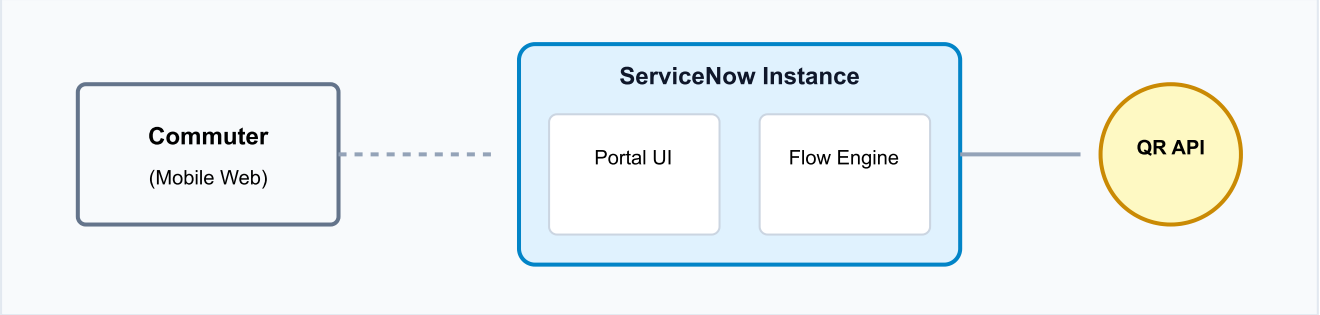
Metro Ticket Booking Digitalization

Phase 3: Project Design & Architecture

1. High-Level Architecture

The architecture leverages ServiceNow's robust native capabilities, separating the presentation layer (Service Portal) from the business logic layer (Flow Designer) and the data layer (custom tables).



2. Data Model & Custom Tables

Table Name	Purpose	Key Fields
`u_metro_station`	Stores all operational stations in the network.	Station Name, Code, Line Color, Status
`u_fare_matrix`	Defines the cost between any two points.	Origin (Ref), Destination (Ref), Base Fare
`u_ticket_booking`	Core transactional table for individual journeys.	Passenger Name, Origin, Dest, QR Code (URL), Status
`u_payment_log`	Audits financial transactions per ticket.	Ticket (Ref), Amount, Transaction ID, Timestamp

3. Process Workflow (Flow Designer)

The backend logic triggers instantly when a commuter submits a booking request from the portal. The automated flow follows these explicit sequential steps:

1. **Trigger:** Record created on `u_ticket_booking` table.
2. **Look Up Record:** Query the `u_fare_matrix` table matching the user's selected Origin and Destination stations to retrieve the base fare.
3. **Update Record:** Multiply the base fare by the passenger count and update the 'Total Cost' field on the ticket record.
4. **REST API Action:** Send a POST request via IntegrationHub to the external QR generation service, passing the unique Ticket Number and validity timestamp.
5. **Send Email/SMS:** Upon receiving the QR image URL from the API, trigger a customized notification directly to the commuter's provided contact details.

4. User Interface (UI) Components

Booking Widget

A clean, mobile-first Catalog Item featuring dynamic reference dropdowns for stations. Includes client scripts to prevent selecting identical origin and destination.

Real-time Fare Calculator

An asynchronous GlideAjax call running on the portal form that fetches and displays the estimated cost before the user submits the transaction.

Ticket View Page

A custom Service Portal page that displays the generated QR code prominently, alongside a countdown timer showing the ticket's active validity window.

Admin Control Panel

A workspace dashboard featuring time-series reports on ticket volumes, revenue generation, and a module to quickly disable stations during maintenance.